Assignment 2 Solutions

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Question 1. Let $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 1 & -2 \end{bmatrix}, \mathbf{B} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{bmatrix}$, calculate $(\mathbf{AB})^{10}$.

Solution

$$\mathbf{AB} = \begin{bmatrix} \frac{3}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{3}{2} \end{bmatrix}, \mathbf{BA} = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$$

$$(\mathbf{AB})^{10} = \mathbf{A}(\mathbf{BA})^9\mathbf{B} = \mathbf{A}\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}^9 \mathbf{B} = \mathbf{A}\begin{bmatrix} 1 & 0 \\ 0 & 512 \end{bmatrix}\mathbf{B}.$$

Therefore, $(\mathbf{AB})^{10} = \begin{bmatrix} 1 & 2 \\ 1 & -2 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & 512 \end{bmatrix}\begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{bmatrix}$

$$= \begin{bmatrix} 1 & 1024 \\ 1 & -1024 \end{bmatrix}\begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{bmatrix} = \begin{bmatrix} \frac{1025}{2} & -\frac{1023}{2} \\ -\frac{1023}{2} & \frac{1025}{2} \end{bmatrix}$$

Question 2. Let $\mathbf{a}_1 = [1, 1]^T, \mathbf{a}_2 = [1, -1]^T$, and matrix $\mathbf{A} == \begin{bmatrix} -2 & 6 \\ 0 & 4 \end{bmatrix}$. If $\mathbf{A} = k_1\mathbf{a}_1\mathbf{a}_1^T + k_2\mathbf{a}_1\mathbf{a}_2^T + k_3\mathbf{a}_2\mathbf{a}_1^T + k_4\mathbf{a}_2\mathbf{a}_2^T$, what are the values of $k_i, i = 1, \ldots, 4$.

Solution

$$\mathbf{a}_1\mathbf{a}_1^T = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \mathbf{a}_1\mathbf{a}_2^T = \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix},$$

$$\mathbf{a}_2\mathbf{a}_1^T = \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix}, \mathbf{a}_2\mathbf{a}_2^T = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}.$$

$$k_1\mathbf{a}_1\mathbf{a}_1^T + k_2\mathbf{a}_1\mathbf{a}_2^T + k_3\mathbf{a}_2\mathbf{a}_1^T + k_4\mathbf{a}_2\mathbf{a}_2^T = \begin{bmatrix} -2 & 6 \\ 0 & 4 \end{bmatrix} \Leftrightarrow$$

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix} \begin{bmatrix} k_1 \\ k_2 \\ k_3 \\ k_4 \end{bmatrix} = \begin{bmatrix} -2 \\ 6 \\ 0 \\ 4 \end{bmatrix}.$$

The augmented matrix $\left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & -2 \\ 1 & -1 & 1 & -1 & 6 \\ 1 & 1 & -1 & -1 & 0 \\ 1 & -1 & -1 & 1 & 4 \end{array}\right] \xrightarrow{RREF} \left[\begin{array}{cccc|c} 1 & 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0 & -3 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & -1 \end{array}\right]$

$\Rightarrow k_1 = 2, k_2 = -3, k_3 = 0, k_4 = -1.$

Question 3. Consider a 4×5 matrix $\mathbf{A} = [\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3, \mathbf{a}_4, \mathbf{a}_5]$. The reduced row-echelon form of $\mathbf{A}$ is

$$\mathbf{U} = \begin{bmatrix} 1 & 0 & 2 & 0 & -1 \\ 0 & 1 & 3 & 0 & -2 \\ 0 & 0 & 0 & 1 & 5 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}. \tag{1}$$

Moreover, $\mathbf{x}_0 = (3, 2, 0, 2, 0)^T$ is a solution to $\mathbf{Ax} = \mathbf{b}$ where $\mathbf{b} = (0, 5, 3, 4)^T$.

(a) Find the solution set of $\mathbf{Ax} = \mathbf{b}$.

(b) Recover $\mathbf{A}$ if we already know that $\mathbf{a}_1 = (2, 1, -3, -2)^T$ and $\mathbf{a}_2 = (-1, 2, 3, 1)^T$.

Solution

(a) $\mathbf{Ux} = \mathbf{0} \Rightarrow \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = x_3 \begin{bmatrix} -2 \\ -3 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} 1 \\ 2 \\ 0 \\ -5 \\ 1 \end{bmatrix} = x_3\mathbf{c}_1 + x_5\mathbf{c}_2.$

Therefore, the solution set is $\mathbf{x}_0 + \mathsf{span}\{\mathbf{c}_1, \mathbf{c}_2\}$.

(b) Denote $\mathbf{U} = [\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \mathbf{u}_4, \mathbf{u}_5]$. From $\mathbf{U}$ we know

$$\mathbf{u}_3 = 2\mathbf{u}_1 + 3\mathbf{u}_2 \tag{2a}$$

$$\mathbf{u}_5 = -\mathbf{u}_1 - 2\mathbf{u}_2 + 5\mathbf{u}_4 \tag{2b}$$

$\Rightarrow$

$$\mathbf{a}_3 = 2\mathbf{a}_1 + 3\mathbf{a}_2 = [1, 8, 3, -1]^T \tag{3a}$$

$$\mathbf{a}_5 = -\mathbf{a}_1 - 2\mathbf{a}_2 + 5\mathbf{a}_4 = [0, -5, -3, 0]^T + 5\mathbf{a}_4 \tag{3b}$$

where $\mathbf{a}_4 = [a_{41}, a_{42}, a_{43}, a_{44}]^T$. Substitute the updated $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3, \mathbf{a}_4, \mathbf{a}_5$ into

$$[\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3, \mathbf{a}_4, \mathbf{a}_5]\mathbf{x}_0 = \mathbf{b} \tag{4}$$

$\Rightarrow a_{41} = -2, a_{42} = -1, a_{43} = 3, a_{44} = 4.$ and

$$\mathbf{a}_5 = [-10, -10, 12, 20]^T \tag{5}$$

Question 4. Compute the matrix-matrix multiplication

(a) $diag(1, -2, 4) \begin{bmatrix} 3 & 4 & 1 \\ -2 & 0 & 1 \\ 1 & 2 & 2 \end{bmatrix}$

(b) $\begin{bmatrix} 3 & 4 & 1 \\ -2 & 0 & 1 \\ 1 & 2 & 2 \end{bmatrix} diag(1, -2, 4).$

Solution

(a) $diag(1,-2,4)\begin{bmatrix} 3 & 4 & 1 \\ -2 & 0 & 1 \\ 1 & 2 & 2 \end{bmatrix} = \begin{bmatrix} 3 & 4 & 1 \\ 4 & 0 & -2 \\ 4 & 8 & 8 \end{bmatrix}$.

(b) $\begin{bmatrix} 3 & 4 & 1 \\ -2 & 0 & 1 \\ 1 & 2 & 2 \end{bmatrix} diag(1,-2,4) = \begin{bmatrix} 3 & -8 & 4 \\ -2 & 0 & 4 \\ 1 & -4 & 8 \end{bmatrix}$.

Question 5. Let $\mathbf{A} = \frac{1}{2}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$. WHatcompute $\mathbf{A}^2, \mathbf{A}^3$. What will $\mathbf{A}^n$ be?

Solution

$\mathbf{A}^2 = \frac{1}{2}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$, $\mathbf{A}^3 = \mathbf{A}^2\mathbf{A} = \frac{1}{2}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$, so $\mathbf{A}^n = \frac{1}{2}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$

Question 6. Given $\mathbf{A} \in \mathbb{R}^{2\times 2}$. If $\mathbf{A}$ commutes with $\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$, show that if $\mathbf{A} = \begin{bmatrix} a & 0 \\ c & a \end{bmatrix}$ for some a and c.

Solution

Suppose $\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$,

then $\mathbf{A}\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}\mathbf{A} \Leftrightarrow \begin{bmatrix} b & 0 \\ d & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ a & b \end{bmatrix} \Rightarrow b = 0, \quad a = d.$

Therefore, $\mathbf{A}$ has a form $\mathbf{A} = \begin{bmatrix} a & 0 \\ c & a \end{bmatrix}$

Question 7. Compute the following by using block multiplication (all blocks are $k \times k$).

1. $\begin{bmatrix} \mathbf{I} & \mathbf{X} \\ \mathbf{0} & \mathbf{I} \end{bmatrix}\begin{bmatrix} \mathbf{I} & \mathbf{-X} \\ \mathbf{0} & \mathbf{I} \end{bmatrix}$
2. $[\mathbf{I} \quad \mathbf{X}^T][\mathbf{-X} \quad \mathbf{I}]^T$

Solution

1. $$\begin{bmatrix} \mathbf{I} & \mathbf{X} \\ \mathbf{0} & \mathbf{I} \end{bmatrix} \begin{bmatrix} \mathbf{I} & -\mathbf{X} \\ \mathbf{0} & \mathbf{I} \end{bmatrix} = \begin{bmatrix} \mathbf{I}^2 + \mathbf{X0} & -\mathbf{IX} + \mathbf{XI} \\ \mathbf{0I} + \mathbf{I0} & -\mathbf{0X} + \mathbf{I}^2 \end{bmatrix} = \begin{bmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{0} & \mathbf{I} \end{bmatrix} = \mathbf{I}_{2k}$$

2. $$\begin{bmatrix} \mathbf{I} & \mathbf{X}^T \end{bmatrix} \begin{bmatrix} -\mathbf{X} & \mathbf{I} \end{bmatrix}^T = \begin{bmatrix} \mathbf{I} & \mathbf{X}^T \end{bmatrix} \begin{bmatrix} -\mathbf{X}^T \\ \mathbf{I} \end{bmatrix} = -\mathbf{I}\mathbf{X}^T + \mathbf{X}^T\mathbf{I} = \mathbf{O}_k$$

Question 8. Compute $\mathbf{AB}$ by using the following block partitioning.

$$\mathbf{A} = \left[\begin{array}{cc|cc} 2 & 1 & 3 & 2 \\ 1 & 0 & 1 & 2 \\ \hline 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right] \quad \mathbf{B} = \left[\begin{array}{cc|c} 1 & -2 & 0 \\ -1 & 0 & 0 \\ \hline 0 & 5 & 2 \\ -1 & 1 & 0 \end{array}\right]$$

Solution

$$\mathbf{A} = \left[\begin{array}{c|c} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \hline \mathbf{A}_{21} & \mathbf{A}_{22} \end{array}\right] \quad \mathbf{B} = \left[\begin{array}{c|c} \mathbf{B}_{11} & \mathbf{B}_{12} \\ \hline \mathbf{B}_{21} & \mathbf{B}_{22} \end{array}\right]$$

$$\mathbf{AB} = \begin{bmatrix} \mathbf{A}_{11}\mathbf{B}_{11} + \mathbf{A}_{12}\mathbf{B}_{21} & \mathbf{A}_{11}\mathbf{B}_{12} + \mathbf{A}_{12}\mathbf{B}_{22} \\ \mathbf{A}_{21}\mathbf{B}_{11} + \mathbf{A}_{22}\mathbf{B}_{21} & \mathbf{A}_{21}\mathbf{B}_{12} + \mathbf{A}_{22}\mathbf{B}_{22} \end{bmatrix} = \begin{bmatrix} -1 & 13 & 6 \\ -1 & 5 & 2 \\ 0 & 5 & 2 \\ -1 & 1 & 0 \end{bmatrix},$$

Question 9. For any $\mathbf{A} \in \mathbb{R}^{n\times n}, \mathbf{B} \in \mathbb{R}^{n\times n}$, show that $\mathbf{AB} = \mathbf{BA}$ if and only if $(\mathbf{A} + \mathbf{B})(\mathbf{A} - \mathbf{B}) = (\mathbf{A} - \mathbf{B})(\mathbf{A} + \mathbf{B})$.

Solution

We always have $(\mathbf{A} + \mathbf{B})(\mathbf{A} - \mathbf{B}) = \mathbf{A}^2 + \mathbf{BA} - \mathbf{AB} - \mathbf{B}^2$.
If $\mathbf{AB} = \mathbf{BA}$, this gives $(\mathbf{A} + \mathbf{B})(\mathbf{A} - \mathbf{B}) = \mathbf{A}^2 - \mathbf{B}^2$.
Conversely, suppose that $(\mathbf{A} + \mathbf{B})(\mathbf{A} - \mathbf{B}) = \mathbf{A}^2 - \mathbf{B}^2$. Then $\mathbf{A}^2 - \mathbf{B}^2 = \mathbf{A}^2 + \mathbf{BA} - \mathbf{AB} - \mathbf{B}^2$. Hence $\mathbf{0} = \mathbf{BA} - \mathbf{AB} \Rightarrow \mathbf{AB} = \mathbf{BA}$.

Question 10. Find $\mathbf{A}$ if $(\mathbf{A}^T - 2\mathbf{I})^{-1} = \begin{bmatrix} 2 & 1 \\ -1 & 0 \end{bmatrix}$

Solution

We have $\mathbf{A}^T - 2\mathbf{I} = [(\mathbf{A}^T - 2\mathbf{I})^{-1}]^{-1} = \begin{bmatrix} 2 & 1 \\ -1 & 0 \end{bmatrix}^{-1} = \begin{bmatrix} 0 & -1 \\ 1 & 2 \end{bmatrix}$. Hence $\mathbf{A}^T = 2\mathbf{I} + \begin{bmatrix} 0 & -1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ 1 & 4 \end{bmatrix}$, so $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ -1 & 4 \end{bmatrix}$.

Question 11. (a)Let $\mathbf{A}$ be a nonsingular matrix. Show that $\mathbf{A}^{-1}$ is also nonsingular and $(\mathbf{A}^{-1})^{-1} = \mathbf{A}$.

(b)Let $\mathbf{A}$ be a nonsingular matrix. Show that $\mathbf{A}^T$ is also nonsingular and $(\mathbf{A}^T)^{-1} = (\mathbf{A}^{-1})^T$.

Solution

(a) Since $\mathbf{A}$ is a nonsingular matrix,
i.e. $\exists\ \mathbf{A}^{-1}$ such that $\mathbf{A}\mathbf{A}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{I}$.
Thus the inverse of $\mathbf{A}^{-1}$ is $\mathbf{A}$.
(b) $\mathbf{A}^T(\mathbf{A}^{-1})^T = (\mathbf{A}^{-1}\mathbf{A})^T = \mathbf{I}^T = \mathbf{I}$.
and $(\mathbf{A}^{-1})^T\mathbf{A}^T = (\mathbf{A}\mathbf{A}^{-1})^T = \mathbf{I}^T = \mathbf{I}$.

Question 12. Give $\mathbf{A}^{-1} = \begin{bmatrix} 1 & -1 & 3 \\ 2 & 0 & 5 \\ -1 & 1 & 0 \end{bmatrix}$, Find a matrix $\mathbf{B}$ such that $\mathbf{AB} = \begin{bmatrix} 1 & -1 & 2 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}$.

Solution

We want B such that $\mathbf{AB} = \mathbf{P}$, where $\mathbf{P} = \begin{bmatrix} 1 & -1 & 2 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}$. Since $\mathbf{A}^{-1}$ exists left multiply this equation by $\mathbf{A}^{-1}$ to get $\mathbf{B} = \mathbf{A}^{-1}(\mathbf{AB}) = \mathbf{A}^{-1}\mathbf{P} = \begin{bmatrix} 1 & -1 & 3 \\ 2 & 0 & 5 \\ -1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & -1 & 2 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 4 & -2 & 1 \\ 7 & -2 & 4 \\ -1 & 2 & -1 \end{bmatrix}$.

Question 13. Let

$$A = \begin{bmatrix} 1 & 0 & 0 \\ a & 1 & 0 \\ b & c & 1 \end{bmatrix}.$$

(a) Write the elementary matrices $\mathbf{E}_1$, $\mathbf{E}_2$ and $\mathbf{E}_3$ from the given $\mathbf{A}$ under the condition that $\mathbf{E}_1$, $\mathbf{E}_2$ and $\mathbf{E}_3$ contain $-a, -b$ and $-c$ respectively.
(b) Multiply $\mathbf{E}_3\mathbf{E}_2\mathbf{E}_1$ to find the single matrix $\mathbf{E}$ that produces $\mathbf{EA} = \mathbf{I}$.
(c) Multiply $\mathbf{E}_1^{-1}\mathbf{E}_2^{-1}\mathbf{E}_3^{-1}$ to bring back $\mathbf{A}$. You will see that the multipliers a, b, c are mixed up in $\mathbf{E}$ but perfect in $\mathbf{A}$.

Solution

(a)

$$\mathbf{E}_1 = \begin{bmatrix} 1 & 0 & 0 \\ -a & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \mathbf{E}_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -b & 0 & 1 \end{bmatrix}, \mathbf{E}_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -c & 1 \end{bmatrix}$$

(b)

$$\mathbf{E} = \mathbf{E}_3\mathbf{E}_2\mathbf{E}_1 = \begin{bmatrix} 1 & 0 & 0 \\ -a & 1 & 0 \\ ac - b & -c & 1 \end{bmatrix}$$

(c)

$$\mathbf{A} = \mathbf{E}_1^{-1}\mathbf{E}_2^{-1}\mathbf{E}_3^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ a & 1 & 0 \\ b & c & 1 \end{bmatrix}$$

Question 14. Find an **LU**-factorization for

$$\begin{bmatrix} -1 & -3 & 1 \\ 1 & 4 & 1 \\ 0 & -2 & -4 \end{bmatrix}.$$

Solution

$$\mathbf{U} = \begin{bmatrix} -1 & -3 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}, \mathbf{L} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -2 & 1 \end{bmatrix}.$$

Question 15. Find a permutation matrix $\mathbf{P}$ and an **LU** decomposition of $\mathbf{PA}$ if $\mathbf{A}$ is

$$\begin{bmatrix} 0 & -2 & 2 \\ 1 & 0 & 4 \\ -1 & 2 & 1 \end{bmatrix}.$$

Solution

$$\mathbf{P} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \mathbf{U} = \begin{bmatrix} 1 & 0 & 4 \\ 0 & -2 & 2 \\ 0 & 0 & 7 \end{bmatrix}, \mathbf{L} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & -1 & 1 \end{bmatrix}.$$

Question 16. Solve the system $\boldsymbol{Ax} = \boldsymbol{b}$ by the provided LU decomposition of $\boldsymbol{A}$.

$$\boldsymbol{A} = \begin{bmatrix} 4 & 3 & -5 \\ -4 & -5 & 7 \\ 8 & 6 & -8 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 4 & 3 & -5 \\ 0 & -2 & 2 \\ 0 & 0 & 2 \end{bmatrix}, \boldsymbol{b} = \begin{bmatrix} 2 \\ -4 \\ 6 \end{bmatrix}$$

Solution

First solve the linear system $\mathbf{L}y = \mathbf{b}$, one can get

$$y = \begin{bmatrix} 2 \\ -2 \\ 2 \end{bmatrix}.$$

Then solve the linear system $\mathbf{U}x = y$, one can get

$$x = \begin{bmatrix} 1/4 \\ 2 \\ 1 \end{bmatrix}.$$